

GANGNEUNG, Korea, Republic of

WMO: 471050

Lat: 37.7514N

Lon: 128.8911E

Elev: 89

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.7	21.3	-12.5	2.8	22.7	-8.9	3.4	25.0	21.3	29.5	18.9	29.9	11.6	230	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.5	92.1	74.8	89.4	73.8	86.8	72.6	77.8	86.6	76.5	84.8	75.4	83.5	7.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.5	133.9	82.1	74.2	128.1	81.0	72.9	122.6	80.1	41.4	86.6	40.2	84.9	39.0	83.4	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	14.7	13.1	12.0	95.7	4.7	2.3	8.7	97.4	5.9	98.7	3.3	100.0	-0.1	101.7
			8.9	79.4	3.9	1.8	6.1	80.7	3.8	81.7	1.6	82.7	-1.3	84.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.6	33.9	37.0	44.7	55.5	64.5	70.3	76.9	77.6	69.2	60.8	49.4	38.2
	DBStd	16.39	6.35	6.52	7.56	7.85	7.14	5.85	6.68	5.46	4.72	5.35	7.01	7.04
	HDD50	1560	498	365	202	30	0	0	0	0	0	3	95	367
	HDD65	4245	963	783	630	300	100	14	1	0	11	147	467	829
	CDD50	3983	0	2	38	197	449	609	835	856	577	339	78	3
	CDD65	1190	0	0	1	16	83	172	371	391	138	17	1	0
	CDH74	9415	0	0	2	101	579	1175	3472	3433	613	40	0	0
	CDH80	3237	0	0	0	16	131	323	1422	1220	122	3	0	0
Wind	WSAvg	5.9	7.7	6.9	6.7	6.5	5.6	4.3	4.4	4.3	4.7	5.7	6.5	7.5
Precipitation	PrecAvg	55.2	2.0	2.1	2.8	3.1	3.1	4.4	9.1	10.9	8.7	4.2	3.1	1.5
	PrecMax	82.5	9.8	7.7	6.9	9.8	10.9	13.5	29.8	44.8	22.0	13.7	13.7	9.2
	PrecMin	36.4	0.0	0.0	0.3	0.0	0.1	0.6	2.3	2.2	0.0	0.1	0.0	0.0
	PrecStd	11.0	2.0	1.9	1.7	2.2	2.2	2.8	5.3	7.0	5.9	3.5	3.0	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.7	59.3	71.2	82.1	88.1	91.1	95.3	95.1	88.1	77.7	70.9	58.6
		MCWB	42.3	46.0	51.5	58.6	62.6	69.0	76.1	77.2	73.8	59.8	56.0	47.0
	2%	DB	48.4	53.9	65.7	76.7	83.6	87.0	92.6	92.0	83.4	74.3	66.7	54.3
		MCWB	38.5	42.1	48.8	56.1	60.4	66.8	75.4	75.8	69.9	60.5	53.0	44.5
	5%	DB	45.4	50.2	61.4	72.9	80.2	83.6	89.9	89.2	80.0	71.9	63.2	50.9
		MCWB	36.4	40.0	46.8	54.2	59.8	66.4	74.5	75.5	68.7	59.6	51.7	41.5
	10%	DB	42.9	47.0	57.5	68.9	76.8	80.7	87.3	86.4	77.2	69.5	60.3	48.1
		MCWB	34.8	37.9	44.4	52.2	58.9	65.9	73.8	74.4	67.4	59.0	50.2	39.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.7	47.7	54.0	63.3	67.1	73.5	79.6	80.0	75.1	66.6	59.0	49.7
		MCDB	50.3	56.6	67.9	76.2	78.7	81.7	90.0	88.9	83.7	71.0	65.8	57.1
	2%	WB	40.2	43.6	50.6	58.6	64.4	71.4	77.7	78.2	72.9	64.5	56.2	45.8
		MCDB	45.6	51.5	62.7	71.5	77.4	79.7	87.3	86.8	79.9	69.6	62.9	52.9
	5%	WB	38.0	41.1	48.0	56.3	62.5	69.6	76.4	77.1	71.0	62.7	53.7	42.6
		MCDB	43.5	48.4	58.1	68.3	73.9	78.2	85.8	84.7	77.2	68.6	60.9	48.8
	10%	WB	35.9	38.9	45.6	54.2	61.0	68.2	75.2	76.1	69.3	60.9	51.5	40.5
		MCDB	41.4	45.4	55.8	65.3	72.2	76.7	84.2	83.4	74.9	67.4	58.2	46.9
Mean Daily Temperature Range	5% DB	MDBR	12.6	13.2	14.4	15.5	14.8	12.2	10.1	10.5	12.0	13.8	13.1	12.7
		MCDBR	14.0	16.3	18.9	19.9	19.9	17.6	13.7	14.2	15.3	16.2	16.0	14.6
		MCWBR	9.7	10.4	10.6	9.8	8.5	7.1	5.2	5.5	7.6	9.1	10.0	10.2
	5% WB	MCDBR	12.4	15.3	17.6	17.9	16.5	14.0	12.2	12.0	12.6	12.7	13.6	13.6
		MCWBR	9.5	10.8	10.9	9.9	8.5	6.7	5.7	5.8	7.0	8.6	9.9	10.6
Clear-Sky Solar Irradiance	taub	0.425	0.491	0.590	0.632	0.637	0.688	0.648	0.575	0.501	0.494	0.457	0.415	
	taud	1.952	1.825	1.666	1.635	1.668	1.629	1.771	1.945	2.072	1.963	1.984	2.007	
	Ebn at Noon	224	224	214	216	219	207	215	228	238	221	212	217	
	Edh at Noon	43	56	73	79	78	81	70	57	48	49	42	39	
All-Sky Solar Radiation	RadAvg	763	1008	1303	1555	1725	1636	1360	1354	1236	1080	765	682	
	RadStd	65	97	129	122	168	137	192	170	149	93	99	58	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-4.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (34 neighbors)	N/A	N/A	-1.69	N/A	N/A	N/A	N/A	N/A	+175	+94

Nomenclature: See separate page